

LECTURE 4: CONVENTIONAL ICE

INTRODUCTION

A conventional vehicle is a vehicle that uses an Internal Combustion Engine (ICE) as its primary propulsion source.

The basic idea is:

Chemical energy stored in fuel → combustion → thermal energy → mechanical energy → vehicle motion.

In a conventional ICE vehicle, there is **no electric motor as the primary propulsion unit**. The energy required for propulsion comes mainly from the combustion of fuel inside the engine. Examples include conventional petrol and diesel vehicles.

Working principle of an ICE

During combustion, the fuel-air mixture burns and produces **high-temperature, high-pressure gases**. These gases expand and exert force on components of the engine.

The force can act on:

- Piston
- Turbine blades
- Rotor
- Other mechanical components

The mechanical force produces motion, which can ultimately be transmitted to the vehicle wheels. Therefore:

Combustion → High-pressure gases → Mechanical motion → Wheel motion

Example: piston-based ICE

In a conventional reciprocating engine:

1. Fuel and air enter the cylinder.
2. The mixture is compressed.
3. Combustion takes place.

4. High-pressure gases are produced.
5. The gases push the piston.
6. The piston produces reciprocating motion.
7. The connecting rod and crank mechanism convert it into rotational motion.
8. The rotation is transmitted through the drivetrain to the wheels.

Important vehicle-performance parameters

The first five pages introduce five major performance parameters:

1. Maximum speed
2. Acceleration
3. Gradeability
4. Range
5. Efficiency

A. Maximum Speed

Maximum speed is the highest speed that a vehicle can achieve under specified conditions, typically on a level road.

The notes define it as: Highest speed that can be achieved on a level road.

At maximum speed, acceleration becomes zero: $a = 0$ Therefore: $F_{\text{net}} = 0$

The available tractive force is just sufficient to overcome the resistive forces.

Thus:

$$F_t = F_r + F_a$$

for a level road.

Factors affecting maximum speed

1. Motor power

Higher available propulsion power can allow the vehicle to maintain a higher speed, provided the other components can support it.

2. Battery power

For an EV, the battery must be capable of supplying the required electrical power to the motor and power electronics. A high-power motor is of limited usefulness if the battery cannot supply sufficient power.

3. Aerodynamic drag

Aerodynamic resistance increases rapidly with speed. Therefore, aerodynamic design becomes very important at high vehicle speeds.

4. Gear ratio

The gear ratio determines the relationship between motor speed and wheel speed. It affects:

- Wheel torque
- Wheel speed
- Motor operating point

B. Acceleration

Acceleration is the **ability of a vehicle to increase its speed quickly**.

A common performance specification is:

$$0 - 100 \text{ km/h in 8 seconds}$$

This means the vehicle takes approximately 8 seconds to accelerate from rest to 100 km/h.

Mathematically:

$$a = \frac{dv}{dt}$$

Requirements for good acceleration

The notes identify three important requirements:

1. High motor torque

The motor should provide sufficiently high torque, particularly at low vehicle speeds.

Wheel torque can be related to motor torque approximately as:

$$T_w = T_m G \eta$$

where:

- T_w = wheel torque
- T_m = motor torque
- G = gear ratio
- η = drivetrain efficiency

Higher wheel torque generally means greater available tractive force.

2. Low vehicle weight

From: $a = \frac{F}{M}$ for the same tractive force, reducing vehicle mass increases acceleration.

For example:

$$M \downarrow \Rightarrow a \uparrow$$

This is one reason lightweight vehicle design is important.

3. High battery-current capability

In an EV, high motor torque generally requires substantial electrical current.

Therefore, the battery and power electronics must be capable of supplying the required current.

A simplified motor-power relationship is:

$$P = VI$$

Thus, the battery must be able to deliver sufficient power without excessive voltage drop or overheating.

C. Gradeability

Gradeability is the ability of a vehicle to climb a slope or hill. For example, a road grade of 20% means that the road rises approximately 20 m for every 100 m of horizontal distance.

1. Grade percentage

Grade can be expressed as:

$$\text{Grade} = \frac{\text{vertical rise}}{\text{horizontal distance}} \times 100$$

For a road angle α :

$$\text{Grade} = \tan \alpha \times 100\%$$

For small angles:

$$\sin \alpha \approx \tan \alpha$$

so the approximation can be used for many vehicle calculations.

2. Factors affecting gradeability

Motor torque

Higher available motor torque provides greater tractive effort at the wheels.

Vehicle mass

A heavier vehicle requires more force to climb a slope.

The gravitational component is:

$$F_g = Mg \sin \alpha$$

Therefore:

$$M \uparrow \Rightarrow F_g \uparrow$$

Tyre grip

The vehicle cannot use unlimited motor torque if the tyres cannot transmit that force to the road.

If the tractive force exceeds the available tyre-road adhesion, the driven wheels may slip.

Therefore, gradeability is affected not only by motor capability but also by **tyre-road adhesion**.

D. Range

Range is the distance a vehicle can travel using the available energy in its battery before the battery reaches the specified lower state-of-charge limit.

Maximum distance travelled on one battery charge. For an EV, range is strongly dependent on the available battery energy and the energy consumed per kilometre.

A simplified relationship is:

$$\text{Range} \approx \frac{\text{Usable battery energy}}{\text{Energy consumption per unit distance}}$$

For example, if:

- usable battery energy = 60 kWh
- energy consumption = 15 kWh/100 km

then:

$$\text{Range} = \frac{60}{15} \times 100$$

$$\text{Range} = 400 \text{ km}$$

This is a simplified calculation; actual range depends on operating conditions.

Factors affecting range

1. Battery capacity

Higher usable battery energy generally provides greater range.

$$E_{\text{battery}} \uparrow \Rightarrow \text{Range} \uparrow$$

provided energy consumption remains unchanged.

2. Vehicle speed

At higher speeds, aerodynamic drag increases significantly.

Since aerodynamic drag increases approximately with the square of speed:

$$F_d \propto v^2$$

the power required to overcome aerodynamic drag increases approximately as:

$$P_d \propto v^3$$

Therefore, highway driving at high speed can significantly increase energy consumption.

3. Driving style

Aggressive driving involving:

- Rapid acceleration
- High speed
- Frequent braking

can increase energy consumption. Smooth driving can reduce unnecessary energy losses.

4. Load condition

A vehicle carrying more passengers or cargo has greater mass.

This increases:

- Required acceleration force
- Grade resistance
- Rolling resistance in practical operation

Therefore, additional load can reduce range.

5. Air-conditioning and auxiliary loads

Although not elaborated in the handwritten page, the basic principle of the page's load discussion is that auxiliary electrical loads consume battery energy.

Examples include:

- Air conditioning
- Heating
- Lighting
- Infotainment
- Pumps and fans

Thus:

$$E_{\text{battery}} = E_{\text{traction}} + E_{\text{auxiliary}}$$

and greater auxiliary consumption leaves less energy available for vehicle propulsion.

E. Efficiency

Efficiency indicates how effectively the input energy is converted into useful output energy.

A general definition is:

$$\eta = \frac{\text{Useful output energy}}{\text{Input energy}} \times 100\%$$

For a vehicle, the useful output is ultimately the energy available at the wheels for producing vehicle motion

The fundamentals of vehicle design involve the basic principles of physics, specially the Newton's second law of motion. According to Newton's second law the acceleration of an object is proportional to the net force exerted on it. Hence, an object accelerates when the net force acting on it is not zero. In a vehicle several forces act on it and the net or resultant force governs the motion according to the Newton's second law. The propulsion unit of the vehicle delivers the force necessary to move the vehicle forward. This force of the propulsion unit helps the vehicle to overcome the resisting forces due to gravity, air and tire resistance. The acceleration of the vehicle depends on:

- the power delivered by the propulsion unit
- the road conditions
- the aerodynamics of the vehicle
- the composite mass of the vehicle

GENERAL DESCRIPTION OF VEHICLE MOVEMENT

The vehicle motion can be completely determined by analysing the forces acting on it in the direction of motion. The forces acting on a vehicle, moving up a grade, are shown in **Figure 1**.

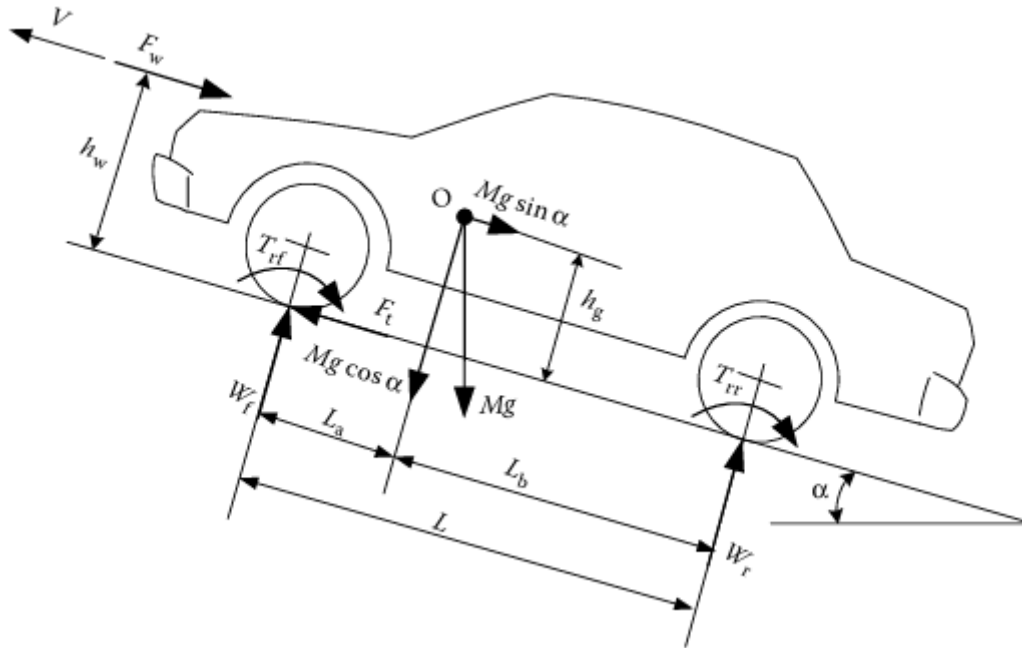


Figure 1: Forces acting on a vehicle moving uphill.

The tractive force (F_t) in the contact area between the tires of the driven wheels and the road surface propels the vehicle forward. The tractive force (F_t) is produced by the power plant and transferred to the driving wheels via the transmission and the final drive. When the vehicle moves, it encounters a resistive force that tries to retard its motion. The resistive forces are

- Rolling resistance
- Aerodynamic drag
- Uphill resistance

Using the Newton's second law of motion, the vehicle acceleration can be expressed as

$$\frac{dV}{dt} = \frac{\sum F_t - \sum F_r}{\delta M}$$

where V is the speed of the vehicle, F_t is the total tractive effort of the vehicle, F_r is the total resistance, M is the total mass of the vehicle, and δ is the mass factor that equivalently converts the rotational inertias of rotating components into translational mass.

Rolling resistance

The rolling resistance of tires on hard surfaces is primarily caused by hysteresis in the tire material. In Figure 2 a tire at standstill on which a force, P , is acting at its center. The pressure

in the contact area between the tire and ground is distributed symmetrically to the central line and the resultant reaction force, P_z , is aligned to P .

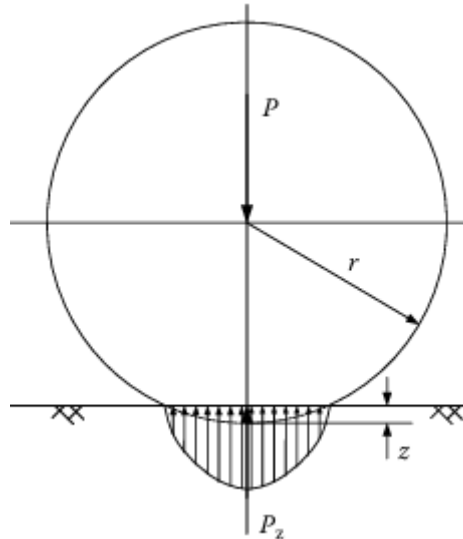


Figure 2: Pressure distribution in contact area

The deformation, z , versus the load, P , in the loading and unloading process is shown in Figure 3. Due to hysteresis in the deformation of rubber material, the load at loading is larger than that at unloading at the same deformation, z , as shown in Figure 3.

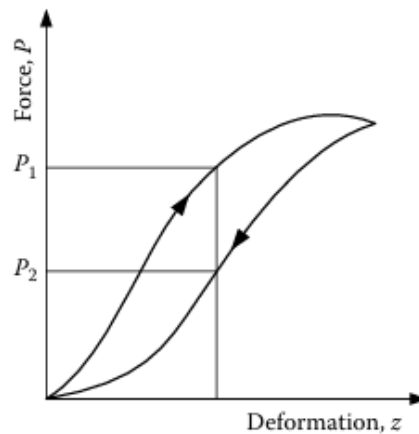


Figure 3: Force acting on a tire versus tire deformation in loading and unloading

When the tire is rolling, as shown in Figure 4a, the leading half of the contact area is loading and the trailing half is unloading. .Consequently, the hysteresis causes an asymmetric distribution of the ground reaction forces. The pressure in the leading half of the contact area is larger than that in the trailing half, as shown in Figure 4a. This phenomenon results in the

ground reaction force shifting forward somewhat. This forwardly shifted ground reaction force, with the normal load acting on the wheel center, creates a moment, which opposes rolling of the wheel. On soft surfaces, the rolling resistance is primarily caused by deformation of the ground surface as shown in Figure 4b. The ground reaction force almost completely shifts to the leading half.

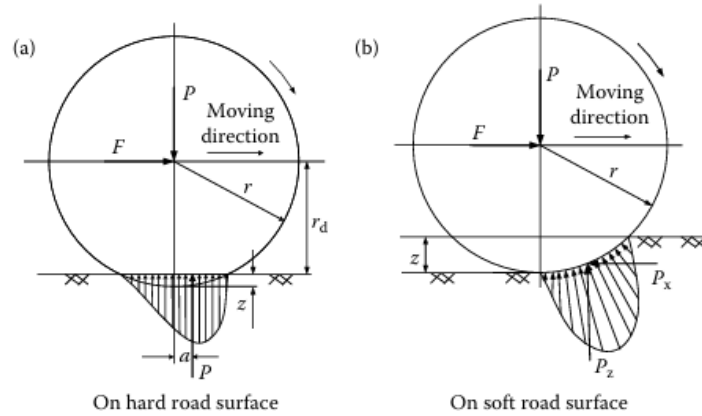


Figure 4: Tire deflection and rolling resistance on a (a) hard and (b) soft road surface.

The moment produced by forward shift of the resultant ground reaction force is called rolling resistance moment (Figure 4a), and can be expressed as

$$T_r = Pa$$

To keep the wheel rolling, a force, F , acting on the center of the wheel is required to balance this rolling resistant moment. This force is expressed as

$$F = \frac{T_r}{r_d} = \frac{Pa}{r_d} = Pf_r$$

Where r_d is the effective radius of the tire and $f_r = a/r_d$ is called the rolling resistance coefficient. In this way, the rolling resistant moment can be equivalently replaced by a horizontal force acting on the wheel center in the opposite movement direction of the wheel. This equivalent force is called rolling resistance with a magnitude of

$$F_r = Pf_r$$

where P is the normal load acting on the center of the rolling wheel. When a vehicle is operated on a slope road, the normal load, P , should be replaced by the component that is perpendicular to the road surface. That is,

$$F_r = P f_r \cos \alpha$$

where α is the road angle (Figure 1)

The rolling resistance coefficient, f_r , is a function of

- tire material,
- tire structure,
- tire temperature,
- tire inflation pressure,
- tread geometry,
- road roughness,
- road material, and
- presence or absence of liquids on the road.

Typical values of rolling resistance coefficients on various roads are given in Table 1. For fuel saving in recent years, low-resistance tires for passenger cars have been developed. Their rolling resistance coefficient is less than 0.01. The values given in Table 1 do not take into account their variations with speed. Based on experimental results, many empirical formulas have been proposed for calculating the rolling resistance on a hard surface.

Table 1: Rolling Resistance Coefficient

Conditions	Rolling Resistance Coefficient
Car tires on a concrete or asphalt road	0.013
Car tires on a rolled gravel road	0.02
Tar macadam road	0.025
Unpaved road	0.05
Field	0.1-0.35
Truck tire on a concrete or asphalt road	0.006-0.01
Wheel on iron rail	0.001-0.002

For example, the rolling resistance coefficient of passenger cars on a concrete road may be calculated from the following equation:

$$f_r = f_0 + f_s \left(\frac{V}{100} \right)^{2.5}$$

where V is vehicle speed in km/h , and f_0 and f_s depend on the inflation pressure of the tire.

In vehicle performance calculation, it is sufficient to consider the rolling resistance coefficient as a linear function of speed. For the most common range of inflation pressure, the following equation can be used for a passenger car on a concrete road.

$$f_r = 0.01 \left(1 + \frac{V}{160} \right)$$

This equation predicts the values of f_r with acceptable accuracy for speeds up to 128 km/h.

Aerodynamic drag

A vehicle traveling at a particular speed in air encounters a force resisting its motion. This force is known as aerodynamic drag. It mainly results from two components:

- shape drag
- skin friction

Shape drag: The forward motion of the vehicle pushes the air in front of it. However, the air cannot instantaneously move out of the way and its pressure is thus increased, resulting in high air pressure. In addition, the air behind the vehicle cannot instantaneously fill the space left by the forward motion of the vehicle. This creates a zone of low air pressure. The motion of the vehicle, therefore, creates two zones of pressure that oppose the motion by pushing (high pressure in front) and pulling it backwards (low pressure at the back) as shown in Figure 5.

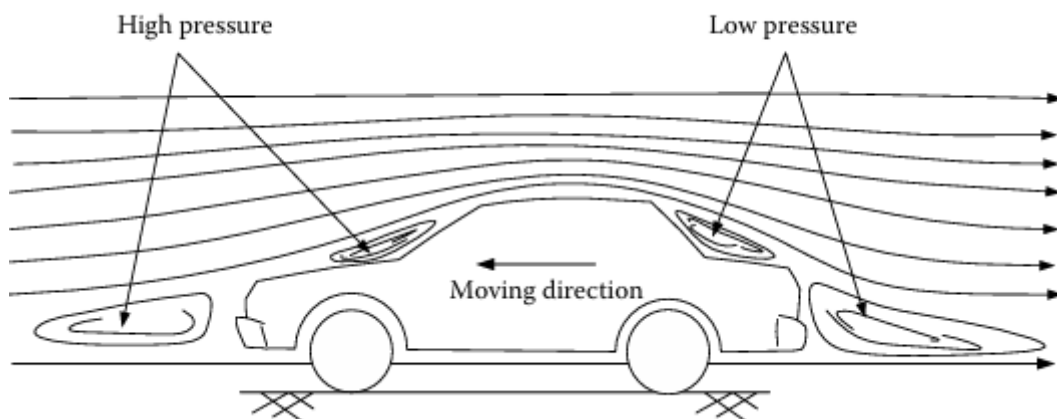


Figure 5: Shape drag

The resulting force on the vehicle is the shape drag. The name “shape drag” comes from the fact that this drag is completely determined by the shape of the vehicle body.

Skin friction: Air close to the skin of the vehicle moves almost at the speed of the vehicle while air away from the vehicle remains still. In between, air molecules move at a wide range of speeds. The difference in speed between two air molecules produces a friction that results in the second component of aerodynamic drag.

Aerodynamic drag is a function of vehicle speed V , vehicle frontal area, A_f , shape of the vehicle body, and air density, ρ :

$$F_w = \frac{1}{2} \rho A_f C_D (V - V_w)^2$$

where C_D is the aerodynamic drag coefficient that characterizes the shape of the vehicle body and V_w is component of the wind speed on the vehicle moving direction, which has a positive sign when this component is in the same direction of the moving vehicle and a negative sign when it is opposite to the vehicle speed. The aerodynamic drag coefficients for typical vehicle body shapes are shown in Figure 6.

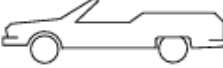
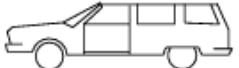
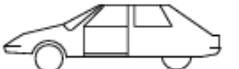
Vehicle type	Coefficient of aerodynamic resistance
 Open convertible	0.5...0.7
 Van body	0.5...0.7
 Ponton body	0.4...0.55
 Wedged-shaped body; headlamps and bumpers are integrated into the body, covered underbody, optimized cooling air flow	0.3...0.4
 Headlamp and all wheels in body, covered underbody	0.2...0.25
 K-shaped (small breakaway section)	0.23
 Optimum streamlined design	0.15...0.20
Trucks, road trains	0.8...1.5
Buses	0.6...0.7
Streamlined buses	0.3...0.4
Motorcycles	0.6...0.7

Figure 6: Indicative drag coefficients for different body shapes.

Grading resistance

When a vehicle goes up or down a slope, its weight produces a component that is always directed in the downward direction, as shown in Figure 7.

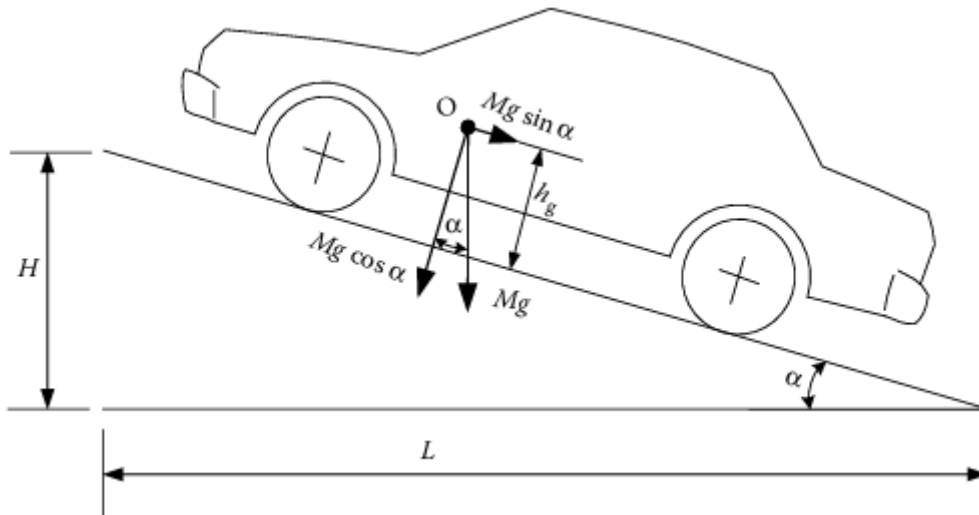


Figure 7: Vehicle climbing a grade

This component either opposes the forward motion (grade climbing) or helps the forward motion (grade descending). In vehicle performance analysis, only uphill operation is considered. This grading force is usually called grading resistance.

Grading resistance, referring to Figure 7, can be expressed

$$F_g = Mg \sin \alpha$$

To simplify the calculation, the road angle, α , is usually replaced by the grade value, when the road angle is small. As shown in Figure 7, grade is defined as

$$i = \frac{H}{L} = \tan \alpha \approx \sin \alpha$$

In some literature, the tire rolling resistance and grading resistance together are called road resistance, which is expressed as

$$F_{rd} = F_f + F_g = Mg(f_r \cos \alpha + \sin \alpha)$$

When the road angle is small, the road resistance can be simplified as

$$F_{rd} = F_f + F_g = Mg(f_r + i)$$

Acceleration resistance

In addition to the driving resistance occurring in steady state motion, inertial forces also occur during acceleration and braking. The total mass of the vehicle and the inertial mass of those rotating parts of the drive accelerated or braked are the factors influencing the resistance to acceleration:

$$F_a = \left(M + \frac{\sum J_{rot}}{r_d^2} \right) \frac{dV}{dt}$$

Where J_{rot} = inertia of rotational components [$kg \times m^2$]

r_d =dynamic radius of the tyre

The rotational component is a function of the gear ratio. The moment of inertia of the rotating drive elements of engine, clutch, gearbox, drive shaft, etc., including all the road wheels are reduced to the driving axle.

The acceleration resistance can be expressed as

$$F_a = \lambda M \frac{dV}{dt}$$

Where λ =rotational inertia constant

Total driving resistance

The traction force (F_t) required at the drive wheels is made up of the driving resistance forces and is defined as

$$F_{resistance} = F_r + F_w + F_g + F_a$$

Substituting the values of all the forces in equation 14, gives

$$F_{resistance} = Mgf_r \cos \alpha + \frac{1}{2} \rho A_f C_D V^2 + Mg \sin \alpha + \lambda M \frac{dV}{dt}$$

The equation may be used to calculate the power required (P_{req}):

$$P_{req} = F_{resistance} V$$

Dynamic equation

In the longitudinal direction, the major external forces acting on a two axle vehicle (Figure 1) include:

- the rolling resistance of the front and rear tires (F_{rf} and F_{rr}), which are represented by rolling resistance moment, T_{rf} and T_{rr}
- the aerodynamic drag (F_w)
- grade climbing resistance (F_g)
- acceleration resistance (F_a)
- tractive effort of the front and rear tires, F_{tf} and F_{tr} . F_{tf} is zero for a rear-wheel-driven vehicle, whereas F_{tr} is zero for a front-wheel-driven vehicle.

The dynamic equation of vehicle motion along the longitudinal direction is given by

$$M \frac{dV}{dt} = (F_{tf} + F_{tr}) - (F_{rf} + F_{rr} + F_w + F_g + F_a)$$

The first term on the right side is the total tractive effort and the second term is the total tractive resistance.

To determine the maximum tractive effort, that the tire ground contact can support, the normal loads on the front and rear axles have to be determined. By summing the moments of all the forces about point R (centre of the tire-ground area), the normal load on the front axle W_f can be determined as

$$W_f = \frac{MgL_b \cos \alpha - (T_{rf} + T_{rr} + F_w h_w + Mgh_g \sin \alpha + Mh_g dV/dt)}{L}$$

Similarly, the normal load acting on the rear axle can be expressed as

$$W_r = \frac{MgL_a \cos \alpha + (T_{rf} + T_{rr} + F_w h_w + Mgh_g \sin \alpha + Mh_g dV/dt)}{L}$$

In case of passenger cars, the height of the centre of application of aerodynamic resistance (h_w) is assumed to be near the height of centre of gravity of the vehicle (h_g). The equation can be simplified as

$$W_f = \frac{L_b}{L} Mg \cos \alpha - \frac{h_g}{L} \left(F_w + F_g + Mg f_r \frac{r_d}{h_g} \cos \alpha + M \frac{dV}{dt} \right)$$

And

$$W_r = \frac{L_a}{L} Mg \cos \alpha + \frac{h_g}{L} \left(F_w + F_g + Mg f_r \frac{r_d}{h_g} \cos \alpha + M \frac{dV}{dt} \right)$$

Where r_d is the effective radius of the wheel. Equations can be rewritten as

$$W_f = \frac{L_b}{L} Mg \cos \alpha - \frac{h_g}{L} \left(F_t - F_r \left(1 - \frac{r_d}{h_g} \right) \right)$$

and

$$W_r = \frac{L_a}{L} Mg \cos \alpha + \frac{h_g}{L} \left(F_t - F_r \left(1 - \frac{r_d}{h_g} \right) \right)$$

The first term on the right hand side of is the static load on the front and the rear axles when the vehicle is at rest on level ground. The second term is the dynamic component of the normal load.

The maximum tractive effort that the tire-ground contact can support (any small amount over this maximum tractive effort will cause the tire to spin on the ground) is usually described by the product of the normal load and the coefficient of road adhesion, μ , or referred to as frictional coefficient. For a front-wheel driven vehicle

$$F_{t \max} = \mu W_f = \mu \left[\frac{L_b}{L} Mg \cos \alpha - \frac{h_g}{L} \left(F_{t \max} - F_r \left(1 - \frac{r_d}{h_g} \right) \right) \right]$$

And

$$F_{t \max} = \frac{\mu Mg \cos \alpha [L_b + f_r(h_g - r_d)]/L}{1 + \mu h_g/L}$$

Where f_r is the coefficient of the rolling resistance. Similarly, for a rear-wheel driven vehicle,

$$F_{t \max} = \mu W_r = \mu \left[\frac{L_a}{L} Mg \cos \alpha + \frac{h_g}{L} \left(F_{t \max} - F_r \left(1 - \frac{r_d}{h_g} \right) \right) \right]$$

And

$$F_{t \max} = \frac{\mu M g \cos \alpha [L_a - f_r (h_g - r_d)] / L}{1 - \mu h_g / L}$$

In vehicle operation, the maximum tractive effort on the driven wheels, transferred from power plant through transmission, should not exceed the maximum values that are limited by the tire–ground cohesion. Otherwise, the driven wheels will spin on the ground, leading to vehicle instability

Road Speed [km/h]	Coefficient of road adhesion for dry roads	Coefficient of road adhesion for wet roads
50	0.85	0.65
90	0.8	0.6
130	0.75	0.55

Adhesion, Dynamic wheel radius and slip

When the tractive effort of a vehicle exceeds the limitation of the maximum tractive effort due to the adhesive capability between the tire and ground, the driven wheels will spin on the ground. Actually, the adhesive capability between the tire and the ground sometimes is the main limitation of vehicle performance. This is especially true when the vehicle drives on wet, icy, snow covered, or soft soil roads. In this case, a tractive torque on the driven wheel would cause the wheel to slip significantly on the ground.

The maximum tractive effort on the driven wheel depends on the longitudinal force that the adhesive capability between the tire and ground can supply, rather than the maximum torque that an engine can supply.

Experimental results show that, on various types of ground, the maximum tractive effort of the drive wheel closely relates to the slipping of the running wheel. This is also true on a good paved, dry road where the slipping is very small due to the elasticity of the tire. The slip, s , of a tire is defined in traction as:

$$S_r = \frac{\omega r - V}{\omega r} \times 100\%$$

where V is the translatory speed of the tire centre, ω is the angular speed of the tire, and r is the rolling radius of the free rolling tire. In traction, the speed, V , is less than ωr ; therefore the slip

of a tire has a positive value between 0 and 1.0. During braking, however, the slip of a tire can be defined as

$$s = \left(1 - \frac{r\omega}{V}\right) \times 100\%$$

which has a positive value between 0 and 1.0, similar to traction.

POWER TRAIN TRACTIVE EFFORT AND VEHICLE SPEED

An automotive power train, as shown in Figure 8, consists of a power plant (engine or electric motor), a clutch in manual transmission or a torque converter in automatic transmission, a gearbox (transmission), final drive, differential, drive shaft, and driven wheels. The torque and rotating speed from the output shaft of the power plant are transmitted to the driven wheels through the clutch or torque converter, gearbox, final drive, differential, and drive shaft. The clutch is used in manual transmission to couple or decouple the gearbox to the power plant.

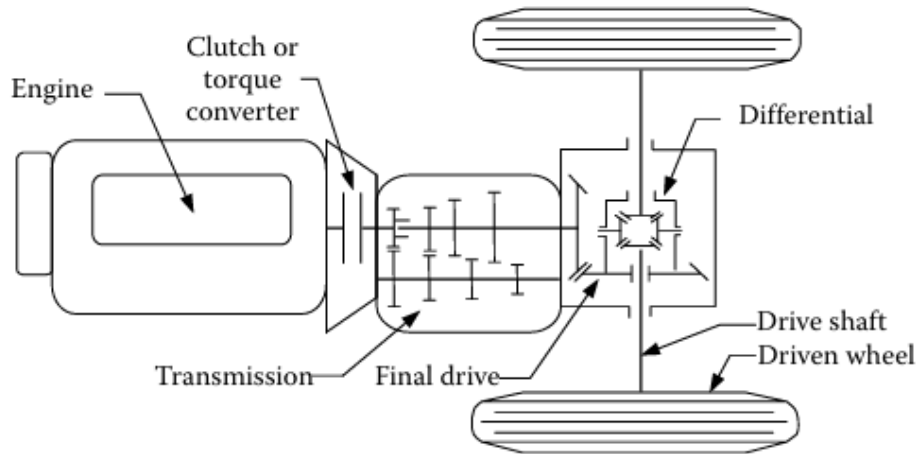


Figure 8: Conceptual illustration of an automobile power train.

The torque converter in automatic transmission is a hydrodynamic device, functioning as the clutch in manual transmission with a continuously variable gear ratio. The gearbox supplies a few gear ratios from its input shaft to its output shaft for the power plant torque–speed profile to match the requirements of the load. The final drive is usually a pair of gears that supply a further speed reduction and distribute the torque to each wheel through the differential. The torque on the driven wheels, transmitted from the power plant, is expressed as

$$T_w = i_g i_0 \eta_t T_p$$

where i_g is the gear ratio of the transmission defined as $i_g = N_{in}/N_{out}$ (N_{in} – input rotating speed, N_{out} – output rotating speed), i_0 is the gear ratio of the final drive, η_t is the efficiency of the driveline from the power plant to the driven wheels, and T_p is the torque output from the power plant.

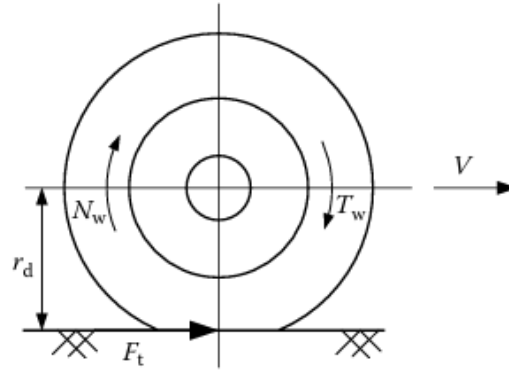


Figure 9: Tractive effort and torque on a driven wheel.

The tractive effort on the driven wheels, as shown in Figure 9, can be expressed as

$$F_t = \frac{T_w}{r_d}$$

Substituting Equation for T_w into Equation for F_t yields the following result:

$$F_t = \frac{T_p i_g i_0 \eta_t}{r_d}$$

The friction in the gear teeth and bearings creates losses in the mechanical gear transmission

$$N_w = \frac{N_p}{i_g i_0}$$

where N_p is the transmission rotating speed (rpm), which is equal to the engine speed in the vehicle with a manual transmission and the turbine speed of a torque converter in the vehicle with an automatic transmission.

The translational speed of the wheel center (vehicle speed) can be expressed as

$$V = \frac{\pi N_w r_d}{30} (m/s)$$

Substituting Equation N_w into Equation, one obtains

$$V = \frac{\pi N_p r_d}{30 i_g i_0} (m/s)$$

VEHICLE POWER PLANT AND TRANSMISSION CHARACTERISTICS

There are two limiting factors to the maximum tractive effort of the vehicle. One is the maximum tractive effort that the tire-ground contact can support and the other is the tractive effort that the maximum torque of the power plant can produce with given drive line gear ratios. The smaller of these two factors will determine the performance potential of the vehicle. For on-road vehicles, the performance is usually limited by the second factor.

Power Plant Characteristics

For vehicular applications, the ideal performance characteristic of a power plant is a constant power output over the full speed range. Consequently, the torque varies with speed hyperbolically as shown in Figure 10. With this ideal profile, the maximum power of the power plant will be available at any vehicle speed, therefore yielding optimal vehicle performance. However, in practice, the torque is constrained to be constant in low speed, so as not to be over the maxima limited by the adhesion between the tire-ground contact area.

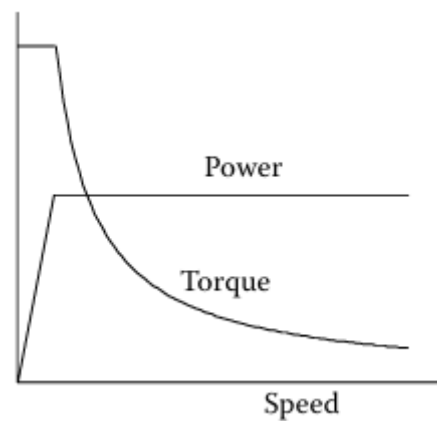


Figure 10: Ideal performance characteristics for a vehicle traction power plant

This constant power characteristic will provide the vehicle with high tractive effort at low speeds where demands for acceleration, drawbar pull, or grade climbing capability are high.

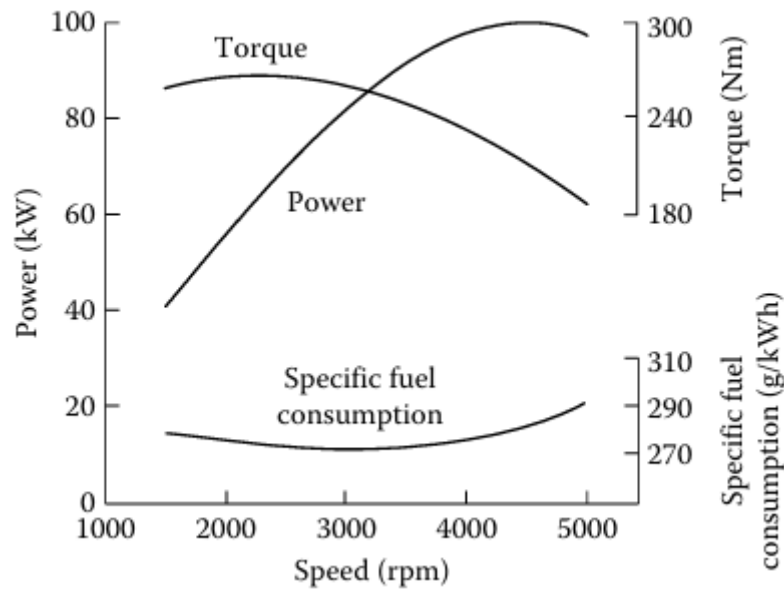


Figure 11: Typical performance characteristics of gasoline engines.

IC engines are the most commonly used power plants for land vehicles to date. Representative characteristics of a gasoline engine in wide open throttle are shown in Figure 11, which has torque–speed characteristics far from the ideal performance characteristic required by traction. It starts operating smoothly at the idle speed. Good combustion quality and maximum torque are reached at an intermediate engine speed. As the speed further increases, the torque decreases due to less air induced into the cylinders, caused by the growing losses in the air-induction manifold and grossing power losses caused by mechanical friction and hydraulic viscosity. Power output, however, increases to its maximum at a certain higher speed. Beyond this speed, It starts operating smoothly at the idle speed. Good combustion quality and maximum torque are reached at an intermediate engine speed. As the speed further increases, the torque decreases due to less air induced into the cylinders, caused by the growing losses in the air-induction manifold and grossing power losses caused by mechanical friction and hydraulic viscosity. Power output, however, increases to its maximum at a certain higher speed. Beyond this speed, the engine power starts declining. In vehicular applications, the maximum permissible speed of the engine is usually set just a little above the speed of the maximum power output. The IC engine has a relatively flat torque–speed profile (as compared with an ideal power plant), as shown in Figure 11. Consequently, a multigear transmission is usually employed to modify it, as shown in Figure 12.

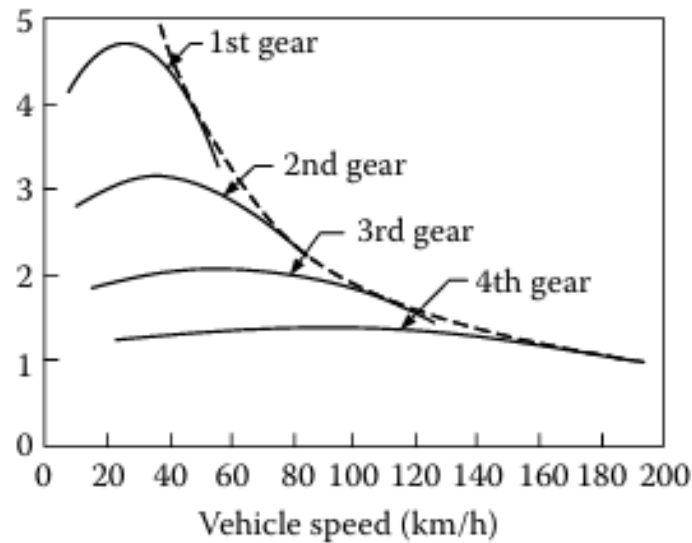


Figure 12: Tractive effort of an IC engine and a multi gear transmission vehicle versus vehicle speed

The electric motor is another candidate as a vehicle power plant, and becoming more and more important with the rapid development of electric, hybrid electric, and fuel cell vehicles. Electric motors with good speed adjustment control usually have a speed–torque characteristic that is much closer to the ideal, as shown in Figure 13.

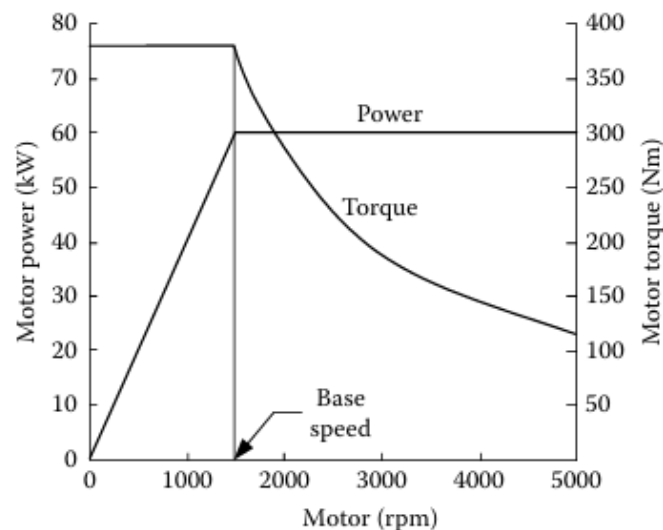


Figure 13: Typical performance characteristics of electric motors for traction.

Generally, the electric motor starts from zero speed. As it increases to its base speed, the voltage increases to its rated value while the flux remains constant. In this speed range of zero to base speed, the electric motor produces a constant torque. Beyond the base speed, the voltage remains constant and the flux is weakened. This results in a constant output power while the

torque declines hyperbolically with speed. Since the speed–torque profile of an electric motor is close to the ideal, a single-gear or double-gear transmission may be employed to meet the vehicle performance requirement, as shown in Figure 14.

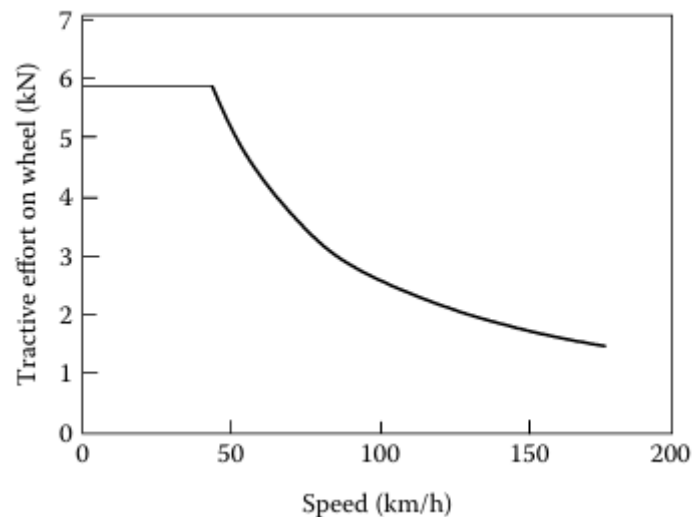


Figure 14: Tractive effort of a single-gear EV versus vehicle speed

Transmission Characteristics

The transmission requirements of a vehicle depend on the characteristics of the power plant and the performance requirements of the vehicle. As mentioned previously, a well-controlled electric machine, such as the power plant of an EV, would not need a multigear transmission. However, an IC engine has to use multigear or continuously varying transmission to multiply its torque at low speed. The term transmission here includes all those systems employed for transmitting the engine power to the drive wheels. For automobile applications, there are usually two basic types of transmission: manual gear transmission and hydrodynamic transmission.

Manual Gear Transmission

Manual gear transmission consists of a clutch, a gearbox, a final drive, and a drive shaft as shown in Figure 8. The final drive has a constant gear ratio. The common practice of requiring direct drive (nonreducing) in the gearbox to be in the highest gear determines this ratio. The gearbox provides a number of gear ratios ranging from three to five for passenger cars and more for heavy commercial vehicles that are powered with gasoline or diesel engines. The maximum speed of the vehicle determines the gear ratio of the highest gear (i.e., the smallest

ratio). On the other hand, the gear ratio of the lowest gear (i.e., the maximum ratio) is determined by the requirement of the maximum tractive effort or the gradeability. Ratios between them should be spaced in such a way that they will provide the tractive effort–speed characteristics as close to the ideal as possible, as shown in Figure 15.

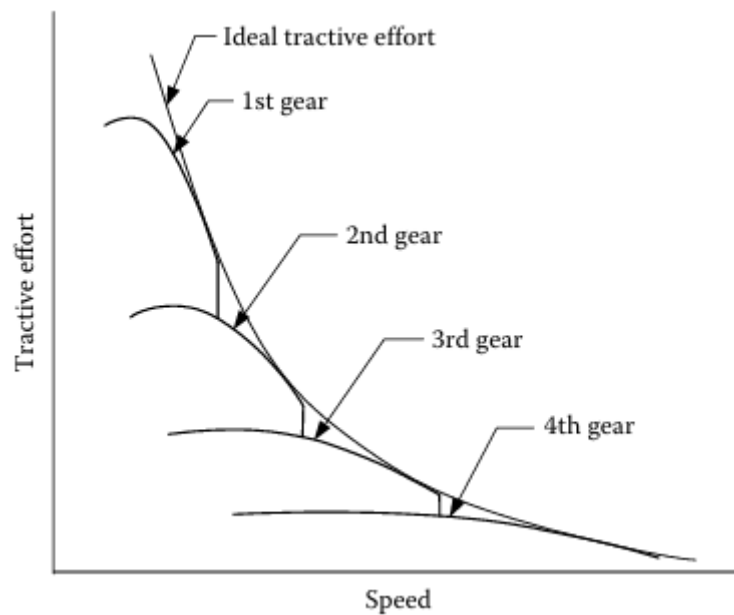


Figure 15: Tractive effort characteristics of a gasoline-engine-powered vehicle

In the first iteration of transmission design, gear ratios between the highest and the lowest gear may be selected in such a way that the engine can operate in the same speed range for all the gears. This approach would benefit the fuel economy and performance of the vehicle. For instance, in normal driving, the proper gear can be selected, according to vehicle speed, to operate the engine in its optimum speed range for fuel-saving purposes. In fast acceleration, the engine can be operated in its speed range with high power output. This approach is depicted in Figure 16.

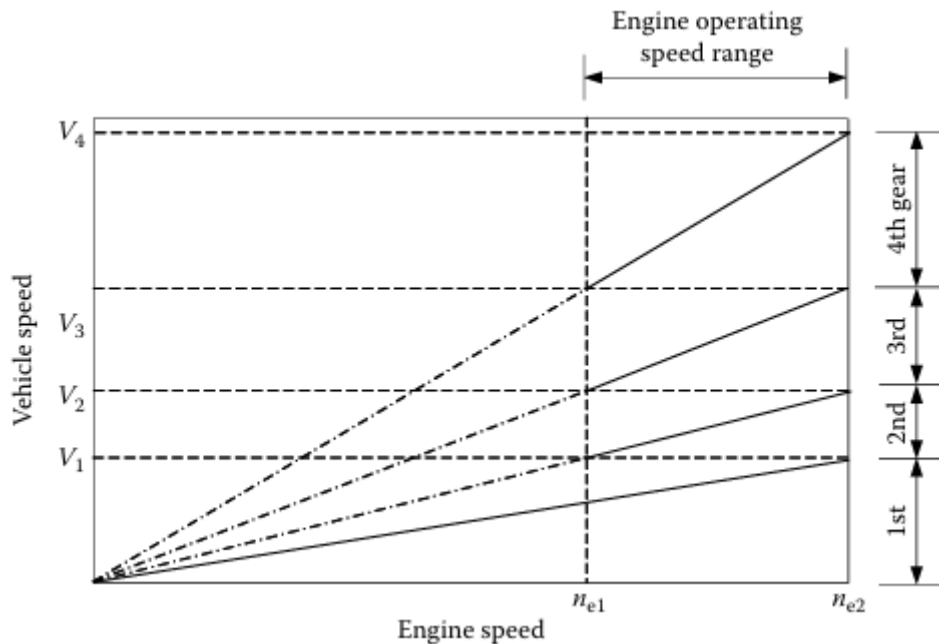


Figure 16: Demonstration of vehicle and speed ranges for each gear.

Hydrodynamic Transmission

Hydrodynamic transmissions use fluid to transmit power in the form of torque and speed. Hydrodynamic transmissions are widely used in passenger cars. They consist of a torque converter and an automatic gearbox. The torque converter consists of at least three rotary elements known as the impeller (pump), the turbine, and the reactor, as shown in Figure 17. The impeller is connected to the engine shaft and the turbine is connected to the output shaft of the converter, which in turn is coupled with the input shaft of the multispeed gearbox. The reactor is coupled to external housing to provide a reaction on the fluid circulating in the converter. The function of the reactor is to enable the turbine to develop output torque higher than the input torque of the converter, thus obtaining torque multiplication. The reactor is usually mounted on a free wheel (one-way clutch) so that when the starting period has been completed and the turbine speed is approaching that of the pump, the reactor is in free rotation. At this point, the converter operates as a fluid coupler with a 1-to-1 ratio of output torque to input torque.

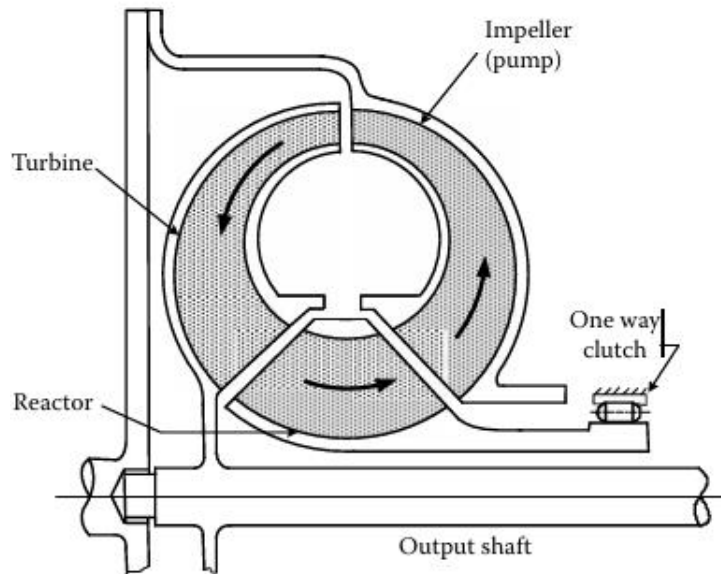


Figure 17: Schematic view of a torque converter.

The major advantages of hydrodynamic transmission may be summarized as follows: •

- When properly matched, the engine will not stall.
- It provides flexible coupling between the engine and the driven wheels.
- Together with a suitably selected multispeed gearbox, it provides torque–speed characteristics that approach the ideal.

The major disadvantages of hydrodynamic transmission are its low efficiency in a stop–go driving pattern and complex structure.

Continuously Variable Transmission

A continuously variable transmission (CVT) has a gear ratio that can be varied continuously within a certain range, thus providing an infinite number of gears. The continuous variation allows for matching virtually any engine speed and torque to any wheel speed and torque. It is therefore possible to achieve an ideal torque–speed profile (constant power profile).

The commonly used CVT in automobiles uses a pulley and belt assembly. One pulley is connected to the engine shaft, while the other is connected to the output shaft. The belt links the two pulleys. The distance between the two half pulleys can be varied, thus varying the effective diameter on which the belt grips. The transmission ratio is a function of the two effective diameters:

$$i_g = \frac{D_2}{D_1}$$

where D_1 and D_2 are the effective diameters of the output pulley and the input pulley, respectively.

Until recently, this implementation was affected by the limited belt–pulley adhesive contact. The design has been improved by the use of metallic belts that provide better solidity and improved contact. Furthermore, an interesting concept has been developed and is being used by Nissan. This concept uses three friction gears: one is connected to the engine shaft, another to the output shaft, and the third grips on the particular profile of the other two gears. It can be rotated to grip on different effective diameters, thus achieving variable gear ratio.

VEHICLE PERFORMANCE

The performance of a vehicle is usually described by its maximum cruising speed, gradeability, and acceleration.

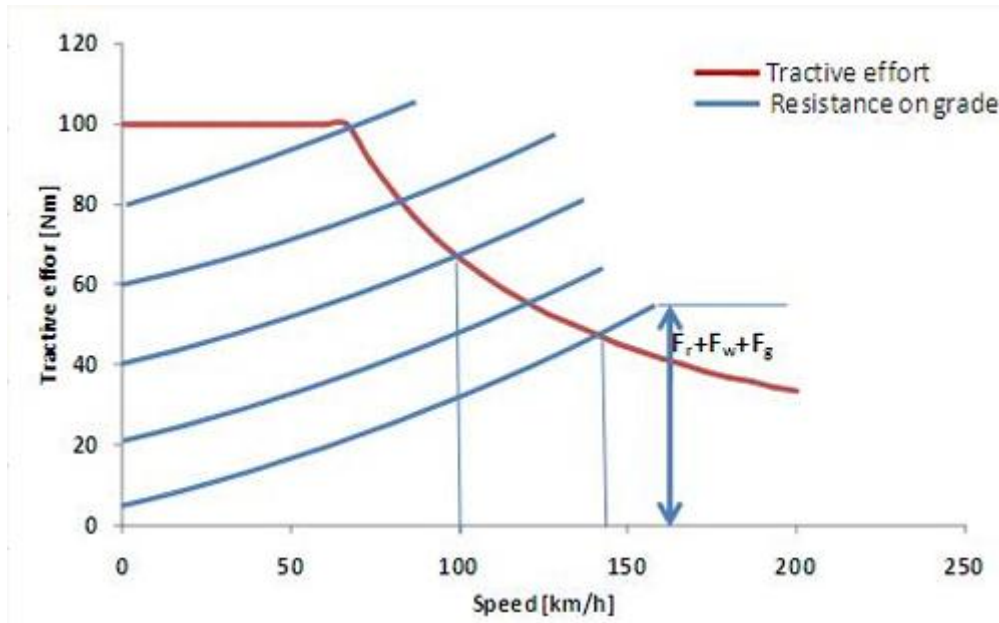
The prediction of vehicle performance is based on the relationship between tractive effort and vehicle speed. For on-road vehicles, it is assumed that the maximum tractive effort is limited by the maximum torque of the power plant, rather than the road adhesion capability.

Maximum Speed of a Vehicle

The maximum speed of a vehicle is defined as the constant cruising speed that the vehicle can develop with full power plant load (full throttle of engine or full power of motor) on a flat road. The maximum speed of the vehicle is determined by the equilibrium between the tractive effort of the vehicle and the resistance or the maximum speed of the power plant and gear ratios of the transmission. The tractive effort and resistance equilibrium can be expressed as

$$\frac{T_p i_g i_0 \eta_t}{r_d} = M g f_r \cos \alpha + \frac{1}{2} \rho_a C_D A_f V^2$$

This equation indicates that the vehicle reaches its maximum speed when the tractive effort, represented by the left-hand-side term in Equation, equals the resistance, represented by the right-hand-side terms. The intersection of the tractive effort curve and the resistance curve is the maximum speed of the vehicle.



It should be noted that for some vehicles, no intersection exists between the tractive effort curve and the resistance curve, because of a large power plant or a large gear ratio. In this case, the maximum speed of the vehicle is determined by the maximum speed of the power plant.

The maximum speed of the vehicle can be obtained by

$$V_{max} = \frac{\pi n_{p \max} r_d}{30 i_0 i_{g \min}} (m/s)$$

where $n_{p \max}$ and $i_{g \min}$ are the maximum speed of the engine (or electric motor) and the minimum gear ratio of the transmission, respectively.

Gradeability

Gradeability is usually defined as the grade (or grade angle) that the vehicle can overcome at a certain constant speed, for instance the grade at a speed of 100km/h(60mph). For heavy commercial vehicles or off-road vehicles, the gradeability is usually defined as the maximum grade or grade angle that the vehicle can overcome in the whole speed range.

When the vehicle drives on a road with relative small grade and constant speed, the tractive effort and resistance equilibrium can be written as

$$\frac{T_p i_g i_0 \eta_t}{r_d} = M g f_r + \frac{1}{2} \rho_a C_D A_f V^2 + M g i$$

Thus,

$$i = \frac{\frac{T_p i_g i_0 \eta_t}{r_d} - M g f_r - \frac{1}{2} \rho_a C_D A_f V^2}{M g} = d - f_r$$

Where

$$d = \frac{F_t - F_w}{M g} = \frac{\frac{T_p i_g i_0 \eta_t}{r_d} - \frac{1}{2} \rho_a C_D A_f V^2}{M g}$$

Is called the performance factor. When the vehicle drives on a road with a large grade, the grade ability of the vehicle can be calculated as

$$\sin \alpha = \frac{d - f_r \sqrt{(1 - d^2 + f_r^2)}}{1 + f_r^2}$$

Acceleration Performance

The acceleration performance of a vehicle is usually described by its acceleration time and distance covered from zero speed to a certain high speed (0 to 96km/h or 60mph, for example) on level ground. Using Newton's second law the acceleration of the vehicle can be written as

$$a = \frac{dV}{dt} = \frac{F_t - F_r - F_w}{M \delta}$$

$$= \frac{\frac{T_p i_g i_0 \eta_t}{r_d} - M g f_r - \frac{1}{2} \rho_a C_D A_f V^2}{M g} = \frac{g}{\delta} (d - f_r)$$

Where δ is called rotational inertia factor, considering the equivalent mass increase due to the angular moments of the rotating components. The mass factor can be written as

$$\delta = 1 + \frac{I_w}{M r_d^2} + \frac{i_0^2 i_g^2 I_p}{M r^2}$$

where I_w is the total angular inertial moment of the wheels and I_p is the total angular inertial moment of the rotating components associated with the power plant. Calculation of the mass factor, δ , requires knowing the values of the mass moments of inertia of all the rotating parts. In the case where these values are not known, the rotational inertia factor, δ , for a passenger car would be estimated using the following empirical relation

$$\delta = 1 + \delta_1 + \delta_2 i_g^2 i_0^2$$

where δ_1 represents the second term on the right-hand side of Equation 2.56, with a reasonable estimate value of 0.04, and δ_2 represents the effect of the power-plant-associated rotating parts, with a reasonable estimate value of 0.0025.

The acceleration time, t_a , and distance, S_a , from low speed V_1 to high-speed V_2 can be written, respectively, as

$$t_a = \int_{V_1}^{V_2} \frac{M\delta}{\frac{T_p i_g i_0 \eta_t}{r_d} - M g f_r - \frac{1}{2} \rho_a C_D A_f V^2} dV$$

And

$$S_a = \int_{V_1}^{V_2} \frac{M\delta V}{\frac{T_p i_g i_0 \eta_t}{r_0} - M g f_r - \frac{1}{2} \rho_a C_D A_f V^2} dV$$